Investigating Feature Selection, Algorithm Performance, and Model Generalizability for Wafer Defect Classification Using Traditional Machine Learning

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Title of thesis	INVESTIGATING FEATURE SELECTION, ALGORITHM PERFORMANCE, AND MODEL GENERALIZABILITY FOR WAFER DEFECT CLASSIFICATION USING TRADITIONAL MACHINE LEARNING

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ABSTRACT

Wafer defect classification is a critical task for yield optimisation in semiconductor manufacturing, yet it remains challenging due to severe class imbalance and limited model generalisability. This thesis investigates the effectiveness of structured feature engineering and systematic feature selection for wafer defect classification using the public WM-811K dataset within a strictly controlled, leakage-free experimental framework. A gatekeeper pipeline was designed in which the test set was fully isolated prior to any preprocessing, class balancing, feature expansion and feature selection. From 66 handcrafted descriptors capturing spatial density, directional, geometric and statistical characteristics, a high-dimensional feature space was generated using interaction-based and polynomial expansions to model non-linear relationships. To mitigate the curse of dimensionality, a hierarchical feature selection strategy combining univariate filtering, Recursive Feature Elimination (RFE), Random Forest importance and L1-regularised logistic regression was employed, where the RFE track produced a compact subset of 25 features. Seven traditional machine learning classifiers were then evaluated under identical conditions using a locked test set and macro-averaged metrics. Logistic Regression trained on the RFE-selected features achieved the highest test Macro F1-score of 0.621, while the Support Vector Machine achieved the highest test accuracy of 87.36% and the smallest generalisation gap, indicating more stable generalisation. The results demonstrate that disciplined feature engineering, strict leakage prevention and controlled model complexity enable traditional machine learning models to achieve strong, interpretable and efficient wafer defect classification performance.

ABSTRAK

Pengelasan kecacatan wafer merupakan satu tugas yang kritikal bagi pengoptimuman hasil (yield) dalam pembuatan semikonduktor, namun ia masih mencabar disebabkan oleh ketidakseimbangan kelas yang ketara serta kebolehgeneralisasian model yang terhad. Tesis ini mengkaji keberkesanan kejuruteraan ciri berstruktur dan pemilihan ciri secara sistematik untuk pengelasan kecacatan wafer menggunakan set data awam WM-811K dalam kerangka eksperimen yang dikawal secara ketat dan bebas kebocoran data, di mana satu saluran pemprosesan gatekeeper direka bentuk dengan pengasingan set ujian sepenuhnya sebelum sebarang prapemprosesan, pengimbangan kelas, pengembangan ciri dan pemilihan ciri dilaksanakan. Bermula daripada 66 deskriptor buatan tangan yang mewakili ciri ketumpatan spatial, arah, geometri dan statistik, satu ruang ciri berdimensi tinggi dijana melalui pengembangan berasaskan interaksi dan polinomial bagi memodelkan hubungan tidak linear. Bagi mengurangkan kesan curse of dimensionality, satu strategi pemilihan ciri berperingkat yang menggabungkan penapisan univariat, Recursive Feature Elimination (RFE), kepentingan ciri berasaskan Random Forest dan regresi logistik berpenalti L1 telah digunakan, di mana laluan RFE menghasilkan satu subset padat sebanyak 25 ciri. Tujuh pengelas pembelajaran mesin tradisional kemudiannya dinilai dalam keadaan yang sama menggunakan set ujian terkunci dan metrik berpurata makro, di mana regresi logistik yang dilatih menggunakan ciri terpilih RFE mencapai skor Macro F1 ujian tertinggi sebanyak 0.621, manakala Support Vector Machine mencapai ketepatan ujian tertinggi sebanyak 87.36% serta jurang generalisasi yang paling kecil, yang menunjukkan kebolehan generalisasi yang lebih stabil. Keputusan ini menunjukkan bahawa kejuruteraan ciri yang berdisiplin, pencegahan kebocoran data yang ketat dan kawalan kerumitan model yang terancang membolehkan model pembelajaran mesin tradisional mencapai prestasi pengelasan kecacatan wafer yang kukuh, boleh ditafsir dan cekap.

DEDICATION

This work is dedicated to my beloved parents and family, whose unwavering support, encouragement, and guidance have been a constant source of strength throughout my academic journey. Their belief in my abilities and sacrifices have made the completion of this research possible.

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LIST OF ABBREVIATIONS

ANOVA — Analysis of Variance
ANN — Artificial Neural Network
CART — Classification and Regression Tree
CNN — Convolutional Neural Network
DT — Decision Tree
GBM — Gradient Boosting Machine
Gini — Gini Impurity Index
IDM — Integrated Device Manufacturer
k-NN — k-Nearest Neighbors
LR — Logistic Regression
L1 — L1 Regularization (Lasso Penalty)
Macro F1 — Macro-Averaged F1-Score
ML — Machine Learning
RFE — Recursive Feature Elimination
RF — Random Forest
SEM — Scanning Electron Microscopy
SMOTE — Synthetic Minority Over-sampling Technique
SVM — Support Vector Machine
WM-811K — Wafer Map Dataset with 811,457 Samples
XGBoost — Extreme Gradient Boosting)

CHAPTER 1

INTRODUCTION

1.1 Overview of the chapter

The introduction chapter of this thesis lays the groundwork for the investigation into the effectiveness of traditional and modern machine learning techniques for wafer defect detection in semiconductor manufacturing. We establish the context of wafer defect detection, its significance in various domains such as yield improvement and reliability assurance, and the challenges associated with accurate and real-time classification. This underscores the need for reliable, objective, and readily deployable methods for identifying wafer defects. This chapter further outlines the motivation behind exploring traditional and modern machine learning methods, focusing on their potential for achieving a balance between accuracy and computational efficiency. The specific research questions we aim to address revolve around identifying the most suitable algorithms and feature sets for robust wafer defect detection. Finally, we delineate the scope of our work, including practical constraints, and provide a roadmap for the structure of the thesis.

1.2 Background about wafer defect

The semiconductor manufacturing industry is a cornerstone of modern technology, producing integrated circuits (ICs) that power a vast array of electronic devices (Chien et al., 2020). The relentless drive towards smaller feature sizes, higher device densities, and improved performance places extreme demands on manufacturing processes (Ma et al., 2023). A critical aspect of this industry is yield management, where the early detection and classification of defects on silicon wafers are paramount (Chien et al., 2020; Ma et al., 2023).

Wafer defects are physical or structural imperfections that occur on semiconductor wafers during various stages of the manufacturing process (Ma et al., 2023). These defects can originate from multiple sources such as contamination, equipment malfunctions, environmental disturbances, or inaccuracies in complex fabrication steps like photolithography and etching (Ma et al., 2023). The semiconductor manufacturing process involves numerous delicate procedures, and any minor deviation or impurity can lead to the formation of defects (Ma et al., 2023). Common types include scratches, particles, edge rings, center clusters, and donut-shaped patterns—each presenting unique challenges in detection and classification (Ma et al., 2023). These defects often appear as anomalies on the wafer surface or as disruptions in the crystalline structure, which can be difficult to detect automatically without advanced image analysis.

Effective wafer defect inspection is critical for ensuring semiconductor device reliability, but traditional approaches have limitations in detecting certain types of defects. Wafer inspection has historically relied on optical microscopy and Scanning Electron Microscopy (SEM) to visually identify surface anomalies such as particles, scratches, pattern errors, and voids (Chien et al., 2020). While effective for many surface-level defects, these techniques may miss sub-surface issues, latent electrical weaknesses, or process-induced stresses that do not produce clear visual features. Such hidden defects may only become evident during electrical testing, either at the wafer probe stage or in early package testing, where voltage and current stresses can expose imperfections through anomalous electrical signatures.

1.3 Impact of Wafer Defects on Semiconductor Manufacturing

1.3.1 Industry Importance and Competitive Implications

Wafer defects are critically important to the semiconductor industry because they directly impact manufacturing efficiency, product reliability, and profitability (Batool et al., 2021; Kim et al., 2023; Ma et al., 2023). In an industry where precision and microscopic accuracy are essential, even the smallest defect can make a semiconductor chip

unusable (Chen et al., 2025; Liu et al., 2024; Purandare et al., 2019). This results not only in increased material waste and rework costs but also in reduced overall production yields (Chien et al., 2013; Gallo et al., 2020; Yang et al., 2017). High defect rates can delay product delivery, reduce customer satisfaction, and increase the risk of returns or failures in end-user applications, especially in sensitive sectors such as automotive electronics, medical devices, and aerospace systems (Fan et al., 2024).

Moreover, the ability to control and minimize wafer defects is a key competitive differentiator among semiconductor manufacturers. Companies that invest in advanced inspection tools, artificial intelligence-driven defect detection, and cleanroom technologies are better positioned to deliver high-performance and reliable chips to market (Huang et al., 2025; Luo et al., 2025; Moon et al., 2025; Sheng et al., 2024; Tsai et al., 2025; Yi et al., 2024; Zhang et al., 2024, 2025). This reliability enhances brand reputation, attracts business from major technology clients, and supports long-term profitability. For foundries, integrated device manufacturers (IDMs), and equipment suppliers alike, wafer defect management is essential not only for ensuring product quality but also for meeting regulatory standards and maintaining global certifications (Chien et al., 2022). Therefore, the semiconductor industry's success hinges significantly on continuous innovation and rigorous control in defect prevention and management.

1.4 Machine Learning for Wafer Defect Detection: Current Approaches and Challenges

The semiconductor industry increasingly depends on automated inspection systems that integrate high-resolution imaging, data processing, and artificial intelligence (AI) to detect and manage wafer defects (Batool et al., 2021; Kim et al., 2023; Ma et al., 2023). These systems leverage pattern recognition and machine learning to classify defects, distinguish between nuisance signals and true yield-impacting flaws, and facilitate timely corrective actions in the production line. Compared to rule-based or manual inspections, machine learning models offer superior adaptability and throughput, which are essential

for modern high-volume manufacturing.

However, several challenges persist in the development of reliable and generalizable machine learning-based defect detection systems. One major issue is limited scalability, whereby models are trained and validated on narrow datasets, leading to poor generalization when applied across different wafer types, imaging systems, or production environments (Batool et al., 2021). In addition, the wide variety of defect shapes and imaging artifacts complicate automated feature extraction. While deep learning models such as convolutional neural networks (CNNs) and transformers can learn features automatically, their performance still depends heavily on model architecture and the diversity of training data.

Algorithm selection and optimization also require substantial effort, as performance varies depending on the specific features used to represent defect patterns (Batool et al., 2021). Furthermore, existing studies rarely conduct systematic evaluations of how different feature sets such as texture, shape, or edge features impact the performance of traditional machine learning algorithms. Lastly, noise introduced by imaging variability and process conditions remains a source of performance degradation, though recent denoising techniques show promise in improving model robustness (Batool et al., 2021).

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1.5 Problem Formulation

1.5.1 Research Gap

Machine learning approaches have been increasingly applied to semiconductor defect detection and wafer map classification (Bae & Kang, 2023; Ferdaus et al., 2022). These studies primarily focus on developing or improving specific models, such as applying supervised contrastive learning for wafer map pattern classification (Bae & Kang, 2023) or investigating the role of activation functions in an online defect classifier for semiconductor inspection (Ferdaus et al., 2022). While these contributions demonstrate improvements in model performance, relatively few studies provide systematic comparisons across multiple machine learning techniques

and feature extraction strategies, which can limit understanding of how different traditional models perform under varying manufacturing conditions and wafer characteristics.As a result, models developed and evaluated on a single dataset may fail to generalize well when applied to other production environments, where variations in defect types, illumination conditions, and imaging resolutions are common (Harada et al., 2019; Hsu & Chia-Yu, 2015).A more comprehensive compari son within a controlled evaluation framework is needed to evaluate the robustness and adaptability of different approaches.

In addition, the selection and combination of features used in wafer defect detection remain underexplored (Altantawy et al., 2024; Fan et al., 2025). Many existing studies apply fixed or manually selected features without systematically examining their effect on model performance across different algorithms (Harada et al., 2019). Since certain models may respond better to specific characteristics such as texture, shape, or edge-related features (Kim et al., 2019), optimizing the feature selection process could enhance both accuracy and computational efficiency. A deeper understanding of the relationship between feature sets and algorithm performance is therefore essential to improve detection reliability and enable scalable solutions.

The absence of broad comparative studies and systematic feature analysis poses a significant challenge in developing reliable and generalizable wafer defect detection systems. Without clear benchmarks and adaptable methodologies, it becomes difficult to determine which techniques are most appropriate for specific industrial applications. This study seeks to fill this gap by providing a detailed comparison of traditional machine learning methods using the WM-811K wafer defect dataset while also analysing the impact of different feature combinations on detection performance.

1.5.2 Problem Statement

This research addresses the limited systematic comparison of machine learning algorithms for wafer defect detection under controlled evaluation settings. There is also a lack of systematic exploration of feature selection methods, which restricts understanding of how different features influence model effectiveness and efficiency. Furthermore, the industrial applicability of existing approaches remains uncertain, as many models are not validated on datasets that reflect diverse real-world manufacturing conditions. This study aims to overcome these challenges by conducting a comparative evaluation of traditional machine learning methods and testing model generalizability using a strictly held-out test set within the WM-811K dataset.

1.6 Research Hypotheses

Based on the problem statement, the following hypotheses are proposed:

1. Traditional machine learning algorithms can effectively classify wafer defects based on wafer inspection data. It is hypothesized that specific patterns in the inspection data can differentiate between defective and non-defective states, and that traditional machine learning algorithms can effectively learn these patterns from labeled data. This hypothesis is based on the premise that defects in semiconductor wafers are reflected in measurable inspection signals.
2. The performance of different machine learning algorithms varies depending on the selected features from wafer inspection data. It is hypothesized that the effectiveness of each algorithm depends on the specific features extracted from the data, and that certain algorithms may be more sensitive to feature characteristics, such as spatial patterns or intensity variations in the inspection images. Optimizing feature selection can therefore significantly impact the performance of different algorithms.
3. Training models on diverse defect categories within the WM-811K dataset improves the generalizability of wafer defect classification models. It is hypothesized that exposing models to a wide variety of defect morphologies, such as cen- ter, edge-ring, scratch, local, and random patterns, during training leads to more robust and stable performance on unseen samples. This diversity of intra-dataset defect patterns reduces the likelihood of overfitting to specific defect structures and enhances the models' ability to generalize across different defect characteristics within the same manufacturing context.

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1.7 Study Objectives

Based on the hypotheses, this study aims to achieve the following objectives:

1. To identify the optimal feature set for each machine learning algorithm by evaluating different feature combinations.
2. To compare the performance of various traditional machine learning algorithms (SVM, DT, RF, KNN, LR, GBM, and XGBoost) for wafer defect classification.
3. To assess model generalization by evaluating performance on a strictly held-out test set containing samples from all defect categories within the WM-811K dataset.

1.8 Scope of the Thesis

The scope of this thesis includes the following components:

1. Preprocessing and structured feature extraction from the publicly available WM-811K wafer defect dataset. This includes cleaning and transforming wafer map representations, followed by the extraction of engineered features such as density-based descriptors, geometric characteristics, statistical measures, and Radon transform features. Feature expansion techniques are also applied to capture higher-order feature interactions.
2. Implementation of a leakage-free machine learning pipeline using a strict stratified train–test split. All preprocessing, feature expansion, scaling, and feature selection procedures are performed exclusively on the training data and subsequently applied to the held-out test set to ensure unbiased evaluation.
3. Training and comparative evaluation of traditional machine learning algorithms, including SVM, Decision Tree, Random Forest, KNN, Logistic Regression, Gradient Boosting, and XGBoost.
4. Systematic analysis of feature selection strategies, including Recursive Feature Elimination (RFE) and model-based feature importance methods, to identify the most discriminative feature subsets for each classifier.
5. Evaluation of model performance using accuracy, macro F1-score, and generaliza

tion gap analysis, together with computational time assessment to examine both predictive effectiveness and computational efficiency.

1.9 Organization Of Thesis

This thesis is organized as follows: Chapter 1 introduces the background, motivation, problem statement, hypotheses, objectives, and scope of the research. It provides a comprehensive overview of the context, rationale, and goals of the study, setting the stage for the subsequent chapters. Chapter 2 reviews the existing literature on wafer defect detection, covering various methods and challenges. This chapter provides a detailed analysis of previous research in this area, identifying the gaps and limitations that motivate the current study. It discusses different approaches to wafer defect detection, including traditional machine learning techniques, deep learning methods, and other image processing approaches.

Chapter 3 details the methodology employed in this study, including data acquisition, preprocessing, feature extraction, model selection, and evaluation metrics. This chapter provides a step-by-step explanation of the methods used in the study, allowing for reproducibility and transparency. It describes the datasets used, the preprocessing techniques applied, the features extracted, the machine learning algorithms employed, and the evaluation metrics used to assess model performance. Chapter 4 presents experimental results, compares the performance of different machine learning algorithms, and analyses the impact of feature selection. This chapter presents the findings of the study, including the performance of different algorithms on various datasets and the impact of different feature combinations. It provides a detailed analysis of the results, highlighting the strengths and weaknesses of different approaches.

Chapter 5 concludes the study by summarising the findings, discussing their implications, and suggesting future research directions. This chapter provides a concise summary of the main contributions of the study and discusses the implications of the findings for the field of wafer defect detection. It also outlines

potential future research directions based on the limitations and open questions arising from this study. The Appendix includes supplementary information such as detailed descriptions of the datasets used, mathematical formulations of the algorithms, and additional experimental results. This section provides supplementary information that may be of interest to readers who wish to delve deeper into the technical details of the study. It includes detailed information about the datasets, algorithms, and experimental setups, providing a comprehensive resource for researchers interested in replicating or extending the study.

CHAPTER 2

LITERATURE REVIEW

2.1 Machine Learning in Wafer Defect classification

Machine learning (ML) has emerged as a valuable approach for detecting defects in semiconductor wafers, where early and accurate identification is critical due to the high cost and sensitivity of fabrication processes (Chen et al., 2002; Chien et al., 2013). Compared to traditional statistical methods, ML techniques offer greater adaptability by learning from large-scale data and generalizing across varying defect types and manu facturing conditions.

In wafer defect classification, machine learning is applied in three main areas. The first is classification, where models categories wafers based on defect types such as scratches, particles, or cracks (Chien et al., 2013). The second is regression, which involves predicting continuous outputs such as defect severity or size based on input features (Demant et al., 2019). The third is anomaly detection, which identifies deviations from normal patterns, enabling the discovery of previously unknown or rare defects in production (Fan et al., 2020).

2.2 Overview of Traditional Machine Learning Algorithms for Wafer Defect classification

This chapter reviews existing literature on the application of traditional machine learning techniques for wafer defect classification. The chapter will examine commonly used ML algorithms, feature engineering approaches, and the characteristics of publicly available wafer defect datasets. The chapter also highlights findings from comparative studies.

2.2.1 Support Vector Machines (SVMs)

Support Vector Machines (SVMs) are widely used in wafer defect classification due to their effectiveness in handling high-dimensional image data and non-linear classification problems. This is particularly relevant in semiconductor manufacturing, where defects captured by scanning electron microscopy (SEM) or wafer inspection systems often involve complex spatial patterns (Gómez-Sirvent et al., 2022; Jeong & Young-Seon, 2017).

In wafer inspection, SVMs have been successfully employed to classify various defect types based on extracted image features. Their ability to map input features into higher-dimensional spaces using kernel functions makes them well-suited for distinguishing subtle defect patterns that traditional classifiers might miss. Studies demonstrate the capability of SVMs to separate defect classes with high accuracy by leveraging texture and geometric features extracted from wafer maps and SEM images (Jeong & Young-Seon, 2017).

SVMs also play an important role in evaluating feature descriptors for defect classi fication. For example, Gomez-Sirvent et al. (2022) utilized SVMs to assess the discriminative power of different feature sets and identified the combination that yielded the highest F1-score. This highlights their utility not only as classifiers but also as tools for guiding feature selection in wafer defect analysis.

Performance metrics reported in the literature support the adoption of SVMs for wafer defect detection. For instance, Gómez-Sirvent et al. (2022) achieved a classifi cation accuracy of 93.13% in identifying wafer map defect patterns, with training and testing completed in under five minutes—demonstrating the practical efficiency of SVMs in industrial applications.

Figure 2.1 illustrates an image set of wafer maps before and after preprocessing using median filtering. This enhancement step is essential in improving defect visibility and suppressing noise, which supports more accurate feature extraction and classification using SVMs. Clearer contrast between defect types such as scratch and center defects can significantly influence the SVM's ability to define effective decision boundaries.

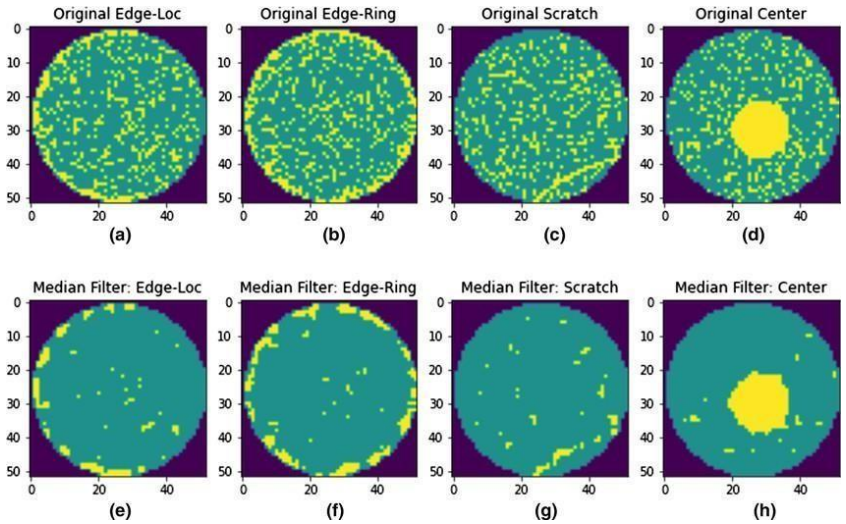

Figure 2.1. Wafer maps before and after median filtering. Preprocessing enhances defect visibility and supports effective feature extraction for SVM classification (T. Kim & Behdinan, 2022).

Overall, the robustness and proven performance of SVMs make them a reliable choice for defect detection in semiconductor wafers, particularly when dealing with complex, high-dimensional image data and the need for fast, accurate classification.

2.2.2 Random Forests (RFs)

Random Forests (RFs) are ensemble learning methods well-regarded for their abil- ity to handle high-dimensional data and mitigate overfitting (Saqlain et al., 2019). These properties are particularly advantageous in semiconductor manufacturing, where datasets often exhibit complex feature interactions and variability across defect types. In wafer defect classification, RFs have demonstrated strong performance in classifying defect patterns and identifying anomalies (Shih et al., 2024). The strength of RFs lies in their capacity to capture intricate relationships between multivariate features, contributing to high classification accuracy in these tasks.

Figure 2.2 presents typical wafer map defect patterns from the WM-811K dataset, including center, edge-ring, scratch, and random defects. These diverse spatial con-

figurations highlight the challenge of separating subtle and overlapping defect types—demonstrating why RFs, with their ensemble structure and feature-based decision boundaries, are well-suited to this problem.

Empirical performance supports RF usage in this domain. (Saqlain et al.2019) reported an accuracy of 95.86 % using an ensemble approach that included RFs (Shih et al.2024) achieved 95.18% in detecting white-pixel anomalies from grinding blade operations. These results underscore the robustness, scalability, and consistent performance of RFs across varied defect scenarios, motivating their adoption in this study.

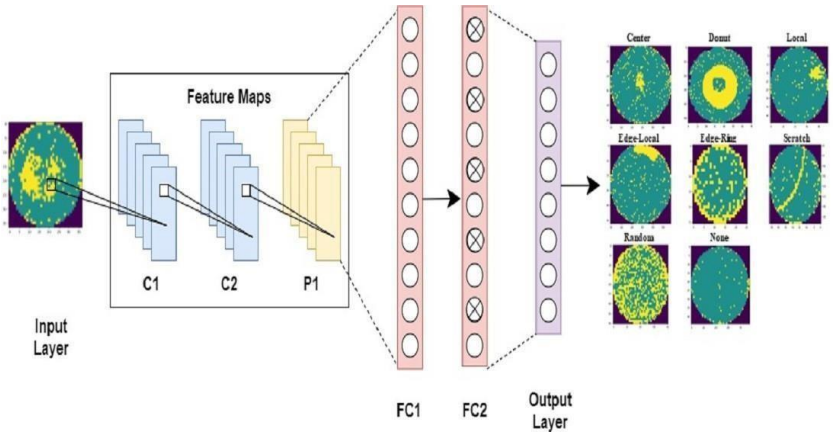

Figure 2.2. Typical wafer map defect patterns from the WM-811K dataset—including center, edge-ring, scratch, and random defects. These diverse spatial patterns illustrate the classification challenges addressed effectively by Random Forest models (Xu et al., 2022).

2.2.3 k-Nearest Neighbors (k-NN)

k-Nearest Neighbors (k-NN) is a non-parametric, instance-based learning algorithm that classifies data points by identifying the majority class among their nearest neighbors in the feature space (Hunt et al., 2000). Its simplicity and interpretability make it a frequently used baseline in classification tasks.

In wafer defect detection, k-NN has been utilized during post-fabrication inspection stages to distinguish between defect types present on wafer surfaces. However, its effectiveness is strongly influenced by the quality and dimensionality of the input features. Wafer map data, often characterized by high dimensionality and noise, can exacerbate k-NN's sensitivity to irrelevant or redundant features.

To address this issue, preprocessing techniques such as median filtering are commonly applied to enhance defect visibility and reduce noise input. As illustrated in Figure 2.3, these techniques improve the clarity of defect patterns, making it easier for k-NN to accurately assign class labels based on neighborhood similarity. Clearer input representations help reduce classification errors caused by irrelevant variations in the raw data.

Kim et al. (2019) conducted a comparative analysis of classification models and found that k-NN's accuracy significantly deteriorates in the presence of irrelevant input variables. This sensitivity was more pronounced than in algorithms such as decision trees, highlighting k-NN's relative vulnerability in high-dimensional settings without proper feature selection or dimensionality reduction.

These findings underscore the importance of robust feature engineering and preprocessing when applying k-NN to wafer defect classification, particularly in environments where noisy or high-dimensional data is common.

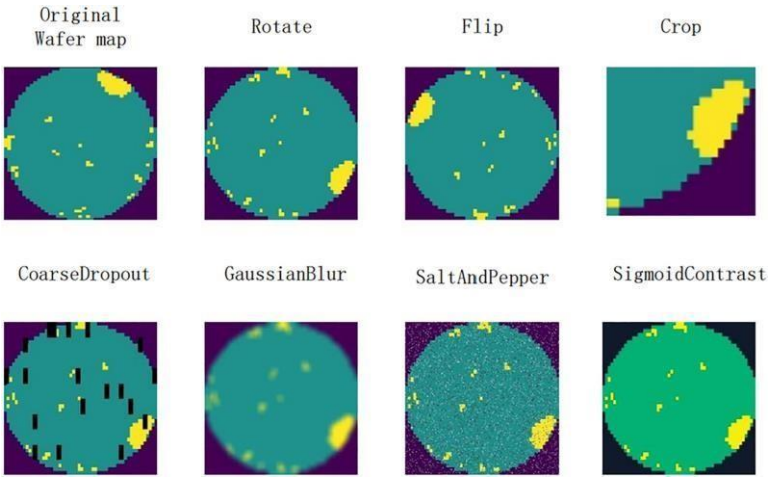

Figure 2.3. Preprocessing effect on wafer defect maps. Median filtering improves visibility of defect patterns by reducing noise, thus enhancing k-NN classification accuracy. Clearer class boundaries result in more reliable distance-based neighbor assignments.

2.2.4 Gradient Boosting Machine (GBM)

Gradient Boosting Machine (GBM) is an ensemble learning technique that builds predictive models by sequentially combining multiple weak learners, typically decision trees, to minimise classification error (Saqlain et al., 2019). Its iterative nature allows it to capture complex, non-linear patterns in data, making it well-suited for high-variability environments such as semiconductor manufacturing.

In wafer defect classification, GBM has demonstrated strong performance in classifying defect patterns. Its capacity to model intricate feature relationships enables the differentiation of various defect types present in wafer inspection data. Saqlain et al. (2019) reported that GBM, when used as part of a voting ensemble, contributed to a classification accuracy of 95.86%, highlighting its effectiveness in real-world applications.

GBM is particularly valued for its robustness to diverse data distributions and its consistent accuracy across different defect categories. These attributes make it a compelling choice for machine learning pipelines aimed at automated defect classification in semiconductor fabrication.

As illustrated in Figure 2.4, the GBM framework enhances prediction by correcting previous errors through sequential model updates, contributing to its strength in handling noisy or imbalanced datasets.

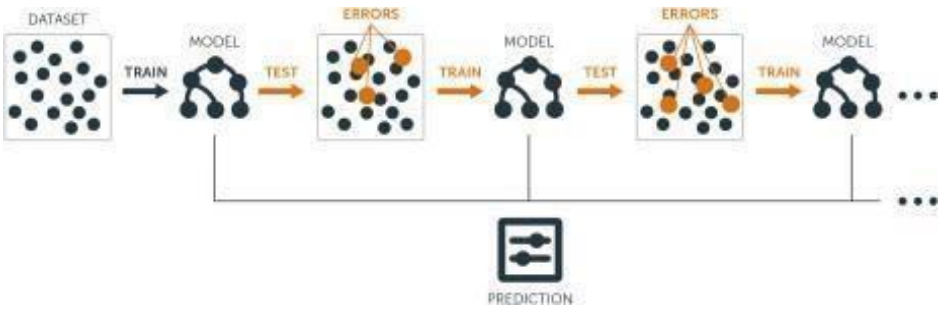

Figure 2.4. Overview of the Gradient Boosting Machine process, where multiple weak learners are combined sequentially to minimize error (Clark, 2020).

2.2.5 XGBoost

XGBoost is a gradient boosting framework that constructs decision trees sequentially, emphasizing the reduction of classification errors and effective handling of class imbalance are the common challenges in wafer defect datasets. Its iterative training process, coupled with regularization techniques, helps prevent overfitting and improves overall classification performance.

Although not traditionally the leading choice in wafer defect detection, XGBoost has gained attention for its effectiveness in ensemble methods tailored to this domain. Its capacity to emphasize misclassified or minority-class samples makes it particularly suitable for the imbalanced and heterogeneous nature of wafer inspection data.

As illustrated in Figure 2.5, the XGBoost pipeline involves stages such as bootstrapping, aggregating, and iterative boosting and each contributing to its high performance on structured datasets (GeeksforGeeks, 2023).

In a study by Choi et al. (2023), XGBoost was employed as a meta-learner in a boosted stacking ensemble model for wafer map defect classification. This ensem-

ble, combining predictions from multiple base classifiers, significantly outperformed individual models in classification accuracy. These findings highlight XGBoost's utility as a core component in hybrid learning architectures for semiconductor defect detection.

While its standalone application remains limited in this field, XGBoost demonstrates strong potential for broader adoption. Future work may focus on applying it directly to wafer image data or integrating it into streamlined ensembles to balance performance and computational efficiency.

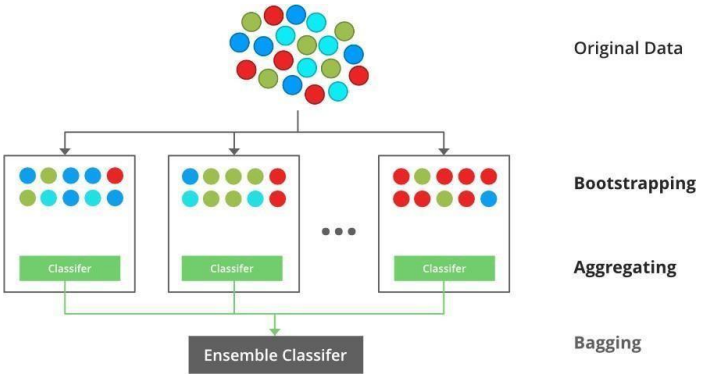

Figure 2.5. Illustration of the XGBoost algorithm pipeline, including data input, bootstrapping, aggregation, and boosting (GeeksforGeeks, 2023).

2.2.6 Decision Trees (DT)

Decision Trees (DT) are non-parametric supervised learning models that classify data by recursively partitioning the feature space based on decision rules derived from input feature values. Each internal node of the tree represents a condition, while each leaf node denotes a class label.

In wafer defect detection, Decision Trees have been employed to classify defect patterns on wafer maps due to their interpretability and ease of implementation. Their transparency allows engineers to trace the decision-making process, which is essential in quality assurance tasks (Hsieh et al., 2004; Kim et al., 2019). For instance, Hunt

(2000) applied DTs as non-parametric classifiers to achieve effective classification of wafer inspection data.

As illustrated in Figure 2.6, the DT classification process structures decisions in a tree format, supporting rule-based classification that can be easily interpreted by human experts (Reader, 2022).

Shibuya et al. (2007) demonstrated the use of DTs to automate rule generation for wafer defect classification using labeled feature data. Kim et al. Kim et al.(2019) further noted DTs' robustness in the presence of irrelevant features, which is advantageous when feature selection is limited. In a related study, Shih et al. (2024) employed CART models to predict white pixel defects caused by grinding blade issues, while Wang (2006) applied DTs using cluster-based features to support quality control decisions.

These studies underscore Decision Trees as practical and interpretable tools for wafer defect classification, particularly valuable in industrial environments where transparency and explainability are critical.

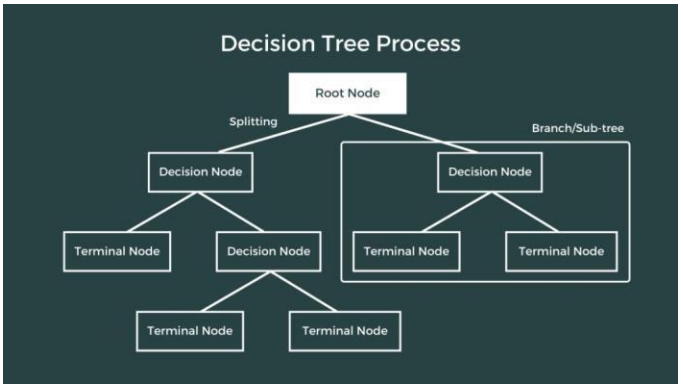

Figure 2.6. Decision Tree classifier process (Reader, 2022).

2.2.7 Logistic Regression (LR)

Logistic Regression (LR) is a linear classification algorithm that estimates the probability of a binary outcome by applying the logistic (sigmoid) function to a linear combi-

nation of input features. It is particularly well-suited for binary classification tasks and often serves as a baseline in many machine learning workflows.

In the context of wafer defect detection, LR has been used to classify wafer map data and assess the separability of defect patterns. Saqlain et al. (2019) included LR in a comparative study, demonstrating its role as a benchmark for evaluating the performance of more complex classifiers.

The appeal of LR lies in its simplicity, computational efficiency, and interpretability. It allows practitioners to examine how individual features contribute to classification outcomes, making it especially useful in the early stages of model development and feature analysis.

As illustrated in Figure 2.7, LR models map input features to a probability score, enabling decision-making based on a threshold. Although limited in capturing non-linear patterns, LR remains a practical starting point for wafer defect classification tasks.

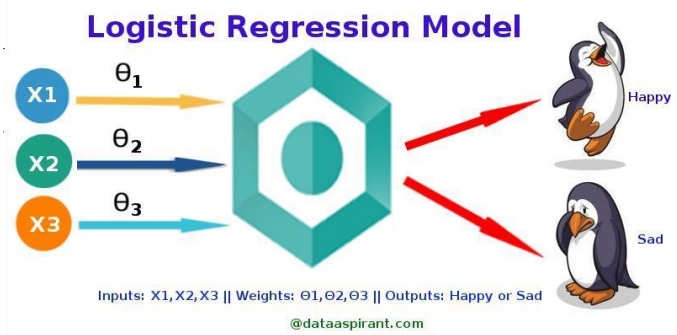

Figure 2.7. Logistic Regression model illustration: features are mapped to probability scores using a sigmoid function (Aspirant, 2020).

2.3 Feature Engineering in Wafer Defect Analysis

Feature engineering plays an important role in wafer defect classification by converting raw inspection data such as spatial signals, defect locations, and measurement values into meaningful inputs for the machine learning models. In semiconductor manufacturing, defects often exhibit spatial or pattern-based characteristics that require careful extrac-

tion of features to enable accurate classification.

Studies such as (Chang et al., 2012) highlight the importance of domain-specific feature extraction techniques, including spatial clustering, geometric descriptors, and statistical summaries derived from wafer map patterns. These engineered features allow algorithms to distinguish between normal and abnormal wafers by capturing subtle variations and repetitive defect signatures across the wafer surface.

Additionally, (Chen et al., 2002) emphasizes transforming raw data into features that capture process-relevant information, which enhances the ability of classification models to detect defect trends and anomalies. Such transformations can include binning spatial regions, quantifying defect density distributions, or encoding shape-based attributes.

Effective feature engineering directly impacts model performance by improving defect detection accuracy and reducing false positives. In wafer defect analysis, this step is indispensable for enabling traditional machine learning models to generalize across varying defect types and process conditions.

Density Features describe how defects are distributed spatially across segmented regions of the wafer map. By dividing the wafer into predefined zones and computing the proportion of defect occurrences in each, density descriptors allow classifiers to distinguish patterns such as edge-ring, center, or localized defects Kulkarni, (2011), Wu et al. (2015).

Radon Transform Features are derived by projecting the wafer map image along multiple angular directions to capture directional information about defect patterns. The resulting sinogram projections provide fixed-length representations of structural characteristics such as scratches or elongated patterns, and are widely used as input features for machine learning models C. Liu and Zhou. (2017), D.-M. Tsai and Hsieh.(2001).

Geometric Features quantify the shape and morphology of defect regions extracted from wafer maps, such as the area, perimeter, major and minor axis lengths, eccentricity, and solidity of connected defect clusters. These features help distinguish between compact clusters and elongated or irregular defect shapes C.-C. Huang and Pan. (2014) and J. Li and Huang.(2018).

Statistical Features summarize global numerical properties of wafer maps, including mean defect density, variance, standard deviation, skewness, and kurtosis of pixel distributions. These statistical descriptors complement spatial and geometric features by characterizing the overall defect load and distribution patterns S. Wang and Liu(2020),Wu et al. (2015).

2.3.1 Feature Selection Approaches in wafer defect

Wrapper Methods involve training a model on different subsets of features and selecting the subset that yields the best predictive performance. In wafer defect classification, this approach helps identify the most relevant features such as spatial, morphological, or statistical attributes that improve classification accuracy. Techniques such as forward selection, backward elimination, and recursive feature elimination fall under this category (Altantawy et al., 2024).

Embedded Methods perform feature selection as part of the model training process. These are particularly useful in wafer defect classification tasks where high-dimensional data is common. Methods like LASSO regression, which includes a regularization term to penalize irrelevant features, and tree-based algorithms, which inherently assess feature importance during model construction, are commonly applied (Fan et al., 2025).

While many studies report the use of these feature types and selection methods individually in wafer defect detection, there is a notable gap in research that systematically combines and evaluates different feature subsets across diverse machine learning models. Exploring these combinations offers a promising direction for improving defect classification performance.

2.4 Dataset Diversity and Model Generalizability

In wafer defect analysis, variability in datasets presents a significant challenge to the generalizability of machine learning (ML) models. These datasets often differ in terms

of input data sources, sensor modalities, environmental conditions, and data quality. Such diversity can lead to inconsistent model performance and difficulties in reproducing results across different settings.

One major source of variation lies in the types of input data used. Some studies utilize image-based data, while others rely on sensor readings, leading to differences in data representation and feature relevance (Fan et al., 2020; Gómez-Sirvent et al., 2022). Sensor modalities also vary, encompassing electrical tests, optical inspections, and scanning electron microscopy, each providing distinct perspectives on wafer characteristics (Chen & Hsiang-Chieh, 2020; Gomez-Sirvent et al., 2022).

In addition, manufacturing environments contribute to data heterogeneity. Variations in process conditions can introduce shifts in data distributions, making it difficult for models trained in one setting to generalise to another (Hatakeyama et al., 2008). Furthermore, data quality issues such as the presence of unlabeled or imbalanced data compound the problem by limiting the model's ability to learn effectively from representative samples.

To address these challenges, several strategies have been proposed. Cross-validation techniques are commonly employed to evaluate model performance across multiple data subsets, providing a more robust assessment of generalizability (Hsu & Chia-Yu, 2015). Domain adaptation methods seek to improve model performance on new, unseen domains by aligning feature representations across different datasets (Ferdaus et al., 2022). Additionally, the use of synthetic data has gained traction as a means of augmenting training datasets, thereby improving model robustness and mitigating issues related to data scarcity or imbalance (Lee et al., 2002).

Despite these advancements, a significant gap remains in current research: many models are tested only in narrow or homogeneous settings. Broader evaluations across diverse datasets are necessary to ensure that ML models for wafer defect detection are not only accurate but also reliably generalized to varied real-world scenarios.

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2.5 Comparative Studies and Benchmarking in ML for Wafer Defect Detection

Existing comparative analyses in wafer defect research have used different datasets for training and testing, which can make direct comparison of machine learning model performance difficult. For example, Bae et al., (2023) evaluated their supervised contrastive learning approach using two different benchmark datasets (WM-811K and MixedWM38) to demonstrate performance gains relative to existing methods. Another major inconsistency lies in the metrics used to evaluate models. Studies frequently rely on varying performance indicators, complicating cross-comparison of results (Choi et al., 2024; Han et al., 2024; Liu et al., 2024). Additionally, many comparisons are based on fixed feature sets, which may not be optimal across all models and can bias the outcomes of such evaluations (Harada et al., 2019).

These comparative studies also tend to focus narrowly on algorithmic performance, without sufficiently exploring how the choice of features or feature combination strategies interacts with different machine learning algorithms. This narrow scope limits understanding of the broader design space in wafer defect detection pipelines.

A significant concern is the lack of standardized benchmarking across feature-algorithm pairings and generalization scenarios. Without common protocols for evaluation, it becomes challenging to identify consistently superior approaches or to draw reliable conclusions about model efficacy. The absence of standardized benchmarking protocols remains a critical gap in current research.

2.5.1 Analysis of Traditional Machine Learning Approaches

Table 2.1 presents a comparative overview of traditional machine learning techniques applied to wafer defect detection. The first column lists the authors and publication years. The second column specifies the algorithms used, such as Subspace Regression, Support Vector Machines (SVM), Random Forest (RF), Gradient Boosting Machines (GBM), Artificial Neural Networks (ANN), and Decision Trees (DT). The third column describes the types of features utilized in each study, which include spatial coordinates, statistical descriptors, shape and location features, texture from local image patches, and process parameters. The fourth column indicates whether the dataset used was publicly available or privately collected. The final column reports performance in terms of classification accuracy, where available.

Saqlain et al., (2019) reported 95.86 percent accuracy using a combination of Random Forest, GBM, and ANN. Shih et al., (2024) demonstrated comparable performance using Decision Trees and Random Forests on private datasets derived from process parameters, with one model achieving 95.18 percent accuracy. Gomez Sirvent et al., (2022) and Li et al., (2009) applied SVMs using texture and spatial/statistical features respectively, both attaining accuracy above 90 percent. Shibuya et al. (2007) employed Decision Trees with separability-based statistical features, reporting an accuracy of 86.6 percent. This comparison highlights that various traditional classifiers can achieve strong performance when paired with suitable features. It also underscores the importance of feature selection and the influence of dataset characteristics on classification accuracy.

Table 2.1. Comparison of Traditional Machine Learning Techniques for Wafer Defect Detection

Author	Algorithm(s)	Features Used	Dataset Type	Performance
Saqlain et al., (2019)	Random Forest, GBM, ANN	Statistical, shape, location (mixed)	Public	95.86% Accuracy
Shih et al., (2024)	Decision Tree	Process parameters (statistical)	Private	95.18% Accuracy
Shih et al., (2024)	Random Forest	Process parameters (statistical)	Private	Not Reported
Gomez Sirvent Et al.,(2022)	Support Vector Machine	Local image patches (texture)	Public	93.13% Accuracy
Li et al.,(2009)	Support Vector Machine	Wafer bin map features (spatial/statistical)	Public	>90% Accuracy
Shibuya et al., (2007)	Decision Tree	Separability measures (statistical)	Public	86.6% Accuracy

CHAPTER 3

METHODOLOGY

This chapter presents the complete technical methodology adopted to develop the wafer defect classification system. The proposed system is based on a structured six-stage machine learning pipeline designed to process the WM-811K wafer map dataset, extract informative high-dimensional descriptors, reduce redundancy through feature selection, and train multiple classification models. A central design principle of this study is the strict prevention of data leakage. To ensure reliable and unbiased performance evaluation, the testing data are fully isolated from all preprocessing, balancing, feature expansion, and selection procedures, in line with best practices in supervised machine learning evaluation (Hastie et al., 2009). Consequently, the reported results reflect true model generalization on previously unseen wafer samples.

The dataset used, WM-811K, consists of anonymized semiconductor manufacturing data collected from a commercial 8-inch wafer fabrication facility. The dataset contains no personally identifiable information (PII), proprietary circuit schematics, or confidential industrial process parameters. The dataset has been publicly released for academic research purposes by the Multimedia Information Retrieval (MIR) Laboratory at National Cheng Kung University, Taiwan. As the data are fully anonymized and publicly available, formal Institutional Review Board (IRB) approval was not required for this study.

Datasets

The dataset used, WM-811K, consists of anonymized semiconductor manufacturing data collected from a commercial 8-inch wafer fabrication facility. The dataset contains no personally identifiable information (PII), proprietary circuit schematics, or confidential industrial process parameters. The dataset has been publicly released for academic research purposes by the Multimedia Information Retrieval (MIR) Laboratory at National Cheng Kung University, Taiwan. As the data are fully anonymized and publicly available, formal Institutional Review Board (IRB) approval was not required for this study.

The WM-811K dataset contains a total of 811,457 wafer maps. However, only 172,950 samples (21.3%) include expert-verified defect pattern labels. The remaining unlabeled samples were excluded to maintain consistency within the supervised learning framework.

Each wafer map is represented as a two-dimensional matrix encoding die-level information, where:

- 0 indicates non-functional die regions,
- 1 represents normal dies, and
- 2 denotes defective dies.

The labeled subset comprises nine classes: eight defect categories (Center, Donut, Edge-Loc, Edge-Ring, Loc, Near-full, Random, Scratch) and one normal class (None). A significant characteristic of the dataset is severe class imbalance, with the None class accounting for approximately 85% of labeled samples, while certain defect categories represent less than 1% of the dataset.

All wafer maps were standardized to a resolution of 64×64 using nearest neighbor interpolation to preserve categorical integrity and ensure consistency across samples.

Experimental Design

The experimental design aimed to evaluate the effectiveness of engineered features for multi-class wafer defect classification using the WM-811K dataset.

A supervised multi-class classification framework was adopted. The labeled dataset was divided using stratified sampling to preserve class distribution across subsets. The partitioning scheme consisted of:

1. Training Set (80%): Used for feature engineering, feature selection (Recursive Feature Elimination and Lasso), scaling parameter estimation, class balancing procedures, and model training.
2. Locked Test Set (20%): Held out from model development and used exclusively for final performance evaluation.

To mitigate class imbalance, a hybrid resampling strategy was applied only to the training data. The majority None class was randomly undersampled, while minority classes were oversampled using the Synthetic Minority Over-sampling Technique (SMOTE). The test set was left unchanged to reflect the natural class distribution.

Primary evaluation metrics included accuracy, macro F1-score, precision, recall, and confusion matrix analysis. This design promotes methodological rigor, reduces the risk of data leakage, and supports fair and reproducible performance evaluation for wafer defect classification.

3.1 Overall Pipeline Structure

The proposed workflow is composed of six sequential and non-overlapping stages, a design commonly adopted to enforce strict data flow constraints and avoid information leakage Kuhn & Johnson., (2013). The first stage performs data ingestion and cleaning, during which invalid or low-quality samples are removed from the dataset. The second stage focuses on fea- ture engineering, where structured numerical descriptors are extracted from each wafer map. The third stage performs data preprocessing, including stratified splitting, feature scaling, and hybrid class balancing. In the fourth stage, feature expansion is applied to model non-linear interactions between attributes. The fifth stage reduces dimensionality through statistical filtering and recursive elimination. Finally, the sixth stage performs model training and hyperparameter optimization using cross-validation.

3.2 Data Ingestion and Initial Cleaning

The pipeline begins by loading the WM-811K wafer map dataset, which is distributed as a serialized Pandas DataFrame and is currently one of the largest publicly wafer defect datasets (Wu et al., 2015). The raw dataset contains more than 811,000 wafer samples, including defective and non-defective patterns. Several data cleaning operations are applied to ensure consistency and reliability.

Samples with missing labels or ambiguous classifications are removed to avoid in- troducing uncertainty during supervised learning. In addition, the Near-full class is excluded, as it is frequently treated in prior studies as non-informative noise rather than a meaningful defect pattern (T. Nakazawa & Kulkarni, 2018). Very small wafer maps with dimensions below 5×5 pixels are discarded because they are susceptible to interpolation artifacts during resizing (LeCun et al., 1998).

To standardize spatial resolution, all wafer maps are resized to a uniform dimension of 64×64 pixels using nearest-neighbor interpolation. This resolution is widely adopted in

wafer map analysis as it preserves defect morphology while maintaining computational efficiency (Yuan et al., 2019). Prior to resizing, a 2×2 median filter is applied to suppress salt-and-pepper noise caused by isolated pixel errors, a standard preprocessing step in binary image denoising (Gonzalez & Woods, 2008).

3.3 Feature Extraction Framework

Feature extraction converts the binary wafer images into structured numerical representations suitable for machine learning models. From each 64×64 wafer map, a total of 66 base features is extracted. These features are grouped into four complementary categories that capture spatial distribution, directional patterns, geometric structure, and statistical characteristics, following established wafer defect analysis frameworks (Y. Li et al., 2018).

3.3.1 Density Features

Thirteen density-based features are computed by dividing each wafer into thirteen spatial zones. These consist of nine internal regions arranged in a 3×3 grid and peripheral edge regions corresponding to the top, bottom, left, and right boundaries. For each zone, the defect density is calculated as the proportion of defective pixels. Such spatial density descriptors have been shown to be effective for identifying location-sensitive defect classes, including Edge-Ring, Center, and Local patterns (Wu et al., 2015).

3.3.2 Radon Transform Features

Directional defect characteristics are captured using Radon transform features. The Radon transforms the wafer image along multiple angular directions between 0° and 180°, enabling the detection of line-oriented structures. In this study, projections are computed at 20 evenly spaced angles. From each projection, the mean and standard deviation are calculated, yielding 40 features in total. These descriptors are particularly effective for identifying linear defects such as Scratch patterns (Y. Liu & Zeng, 2017).

3.3.3 Geometric Features

Geometric and morphological properties are extracted using connected component analysis. The largest connected defect region is identified, and seven geometric descriptors are computed, including area, perimeter, major axis length, minor axis length, eccentricity, solidity, and the number of distinct defect regions. These features are widely used to characterize defect shape complexity and spatial dispersion in wafer map analysis (Haralick et al., 1973).

3.3.4 Statistical Features

Six global statistical features are computed from the pixel distribution of each wafer map, including the mean, variance, standard deviation, skewness, kurtosis, and median. Such global statistics provide a compact representation of overall defect intensity and

noise characteristics and are commonly employed as baseline descriptors in image-based classification tasks (Jain, 1989).

3.4 Data Splitting and Hybrid Balancing

The cleaned dataset is divided into an 80% training set and a 20% locked test set using stratified sampling to preserve class proportions (Kohavi, 1995). The test set is fully isolated prior to any preprocessing or feature selection to prevent data leakage, consistent with recommended evaluation protocols (Hastie et al., 2009).

Due to severe class imbalance in the WM-811K dataset, a hybrid class balancing strategy is applied exclusively to the training data. The majority class is undersampled, while minority classes are oversampled using the Synthetic Minority Over-sampling Technique (SMOTE), a widely adopted method for addressing imbalance in supervised learning (Chawla et al., 2002a).

3.5 Feature Expansion and Selection

3.5.1 Feature Expansion

To capture complex non-linear relationships and interactions among the initial attributes, a comprehensive feature expansion strategy is applied. This expansion is designed to enrich the representational capacity of the feature space while remaining compatible with conventional machine learning models. Two complementary feature generation mechanisms are employed.

First, pairwise arithmetic combinations are constructed by computing the sum and difference between every possible pair of the original 66 features. These derived features allow linear models to explicitly represent additive and subtractive relationships between descriptors, which are not captured by the original feature set alone.

Second, polynomial feature expansion of degree two is applied to generate interaction terms and quadratic components. This includes pairwise interaction features

of the form $x_i x_j$ as well as squared terms x^2, enabling the models to represent non-linear decision boundaries and higher-order dependencies between features. Polynomial expansion is a well-established technique for increasing model expressiveness in high-dimensional learning problems without resorting to fully non-linear classifiers (Friedman, 2001).

Collectively, the arithmetic combinations and polynomial expansion increase the dimensionality of the feature space from 66 to approximately 8,700 features.

3.5.2 Feature Selection Strategy

The substantial increase in dimensionality introduced by feature expansion necessitates an effective feature selection strategy to mitigate the curse of dimensionality, reduce computational complexity, and prevent overfitting. To address these challenges, a hierarchical two-stage feature selection framework is adopted.

3.5.3 Stage 1: Global Pre-Filtering

In the first stage, a univariate Analysis of Variance (ANOVA) F-value test is applied as a global pre-filtering mechanism. Each feature is evaluated independently to as- sess its statistical association with the defect class labels. Features that do not exhibit significant discriminative power are discarded at this stage. This filtering step rapidly reduces the feature space from approximately 8,700 features to a more manageable subset, thereby substantially lowering the computational burden of subsequent selec- tion procedures. ANOVA-based filtering is widely used as an initial screening method in multi-class classification problems due to its simplicity and efficiency (Guyon & Elisseeff, 2003a).

3.5.4 Stage 2: Fine Selection Tracks

In the second stage, three independent feature selection algorithms are applied in parallel to identify compact and discriminative feature subsets. Each track exploits different mathematical principles, providing complementary perspectives on feature relevance and robustness.

Recursive Feature Elimination as tracks 4B is performed using logistic regression as the base estimator. This wrapper-based approach iteratively trains the model, ranks features according to the magnitude of their coefficients, and removes the least informative features at each iteration. The procedure continues until a compact subset of 25 features is retained. Recursive Feature Elimination is particularly effective at capturing feature dependencies and redundancy in high-dimensional spaces (Guyon et al., 2002a).

In parallel, a Random Forest classifier is trained to estimate feature importance as Track 4C based on the mean decrease in impurity. Features contributing most strongly to impurity reduction across the ensemble of decision trees are selected, and the top 25 features are retained. This embedded method is well suited for identifying non-linear and interaction-driven feature relevance.

Finally, feature selection is performed using logistic regression with L_1 regularisation in Track 4D. The L_1 penalty encourages sparsity by forcing the coefficients of less informative features to exactly zero, thereby performing automatic variable selection during model training. Features associated with non-zero coefficients are retained as the selected subset.

Among the three selection tracks, Track 4B (Recursive Feature Elimination) is emphasised in the subsequent analysis because it consistently yielded the most stable performance across multiple classifiers, particularly linear and kernel-based models, while maintaining a compact feature subset and strong generalisation on the locked test set.

3.6 Model Training and Evaluation

Seven machine learning classifiers are trained using the optimised feature set, including Logistic Regression, k-Nearest Neighbors, Decision Tree, Random Forest, Gradient Boosting, Extreme Gradient Boosting (XGBoost), and Support Vector Machines. These models represent a diverse set of linear, non-linear, and ensemble learning paradigms (Bishop, 2006; T. Chen & Guestrin, 2016).

Hyperparameter tuning is performed using three-fold stratified cross-validation on the training data (Kohavi, 1995). Final performance is evaluated on the locked test set using macro-averaged F1-score and macro-averaged recall, which are recommended metrics for imbalanced multi-class classification problems (Saito & Rehmsmeier, 2015). Overfitting is assessed by analyzing the performance gap between training and testing results.

Figure 3.1 illustrates the overall methodological flow of the proposed wafer defect classification framework. where each stage receives inputs exclusively from its immediate predecessor, thereby enforcing unidirectional data flow and preventing information leakage. Raw wafer maps are first subjected to data ingestion and cleaning, followed by structured feature extraction encompassing spatial, directional, geometric, and statistical descriptors. The dataset is then partitioned into stratified training and locked testing subsets prior to any preprocessing or feature selection. All subsequent operations, including feature scaling, hybrid class balancing, polynomial feature expansion, and feature selection, are applied solely to the training data. The final stage involves model training and hyperparameter optimisation using cross-validation, after which performance is evaluated once on the unseen test set. This design ensures that reported results reflect genuine model generalisation rather than optimistic bias introduced by data reuse. The next chapter presents the experimental results obtained from this pipeline and evaluates the effectiveness of the proposed methodology.

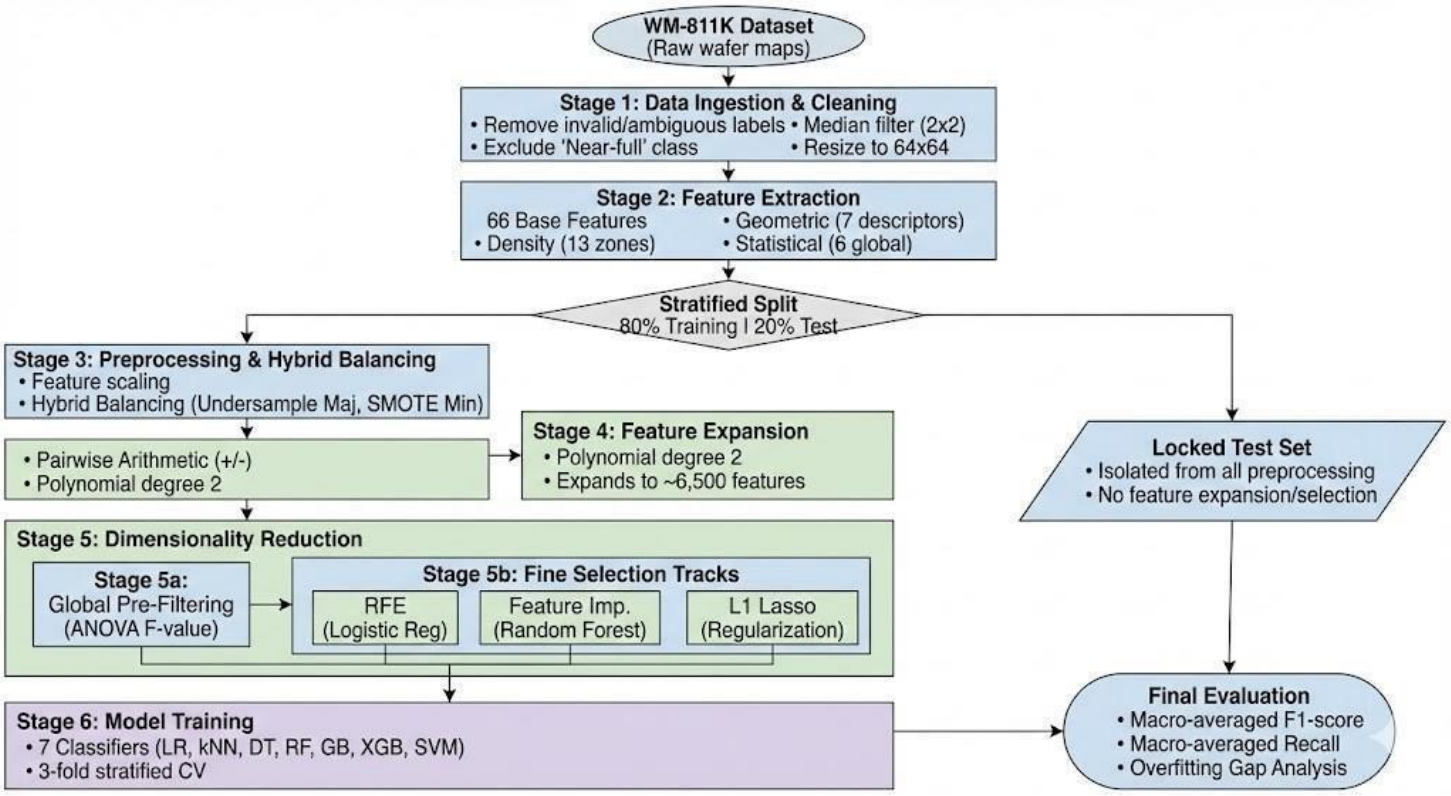

Figure 3.1. End-to-end machine learning pipeline for wafer defect classification on the WM-811K dataset, showing data preprocessing, feature extraction, stratified training–test separation, model training with class balancing, and final evaluation on a locked test set to ensure unbiased performance assessment.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Overview

This chapter details the empirical findings derived from the implementation of the automated machine learning pipeline methodology defined in Chapter 3. The experimental evaluation presented herein serves to validate the proposed leak-proof architectural de- sign and systematically addresses the three specific study objectives outlined in Section 1.7

The structure of the analysis mirrors the sequential stages of the methodological framework to provide a coherent narrative. The evaluation commences in Section 4.2, which addresses the first research objective regarding the identification of an optimal feature set. This section evaluates the efficacy of the feature engineering strategies proposed in the methodology, specifically quantifying the impact of the Feature Combination stage. The analysis determines whether the reduction from a high-dimensional interaction space to a compact feature subset successfully optimizes model convergence while preserving discriminative information.

Fulfilling the second research objective, Section 4.3 benchmarks the performance of the seven machine learning algorithms including Support Vector Machines (SVM), Random Forests, and Gradient Boosting configured during the Model Tuning phase. This comparative study identifies the optimal algorithmic paradigm for wafer defect classification, validating the selection of candidate models discussed in the experimental design and highlighting the trade-offs between computational complexity and classification accuracy.

Finally, Section 4.4 addresses the third objective by assessing the robustness and generalizability of the trained models on the independent Test Set. This analysis directly validates the data partitioning strategy established in the Preprocessing stage, utilizing

the Overfit Gap metric to confirm the system's reliability on unseen production data. Crucially, all results reported in this chapter are derived exclusively from this independent partition, ensuring that the findings reflect true generalization capability free from the data leakage bias often associated with synthetic oversampling techniques like SMOTE (Chawla et al., 2002b).

Collectively, this chapter not only reports experimental performance but also interprets the observed behaviors in relation to defect morphology, feature representation, and model generalization. This dual emphasis ensures that the results are evaluated not only in terms of numerical performance, but also with respect to their physical plausibility and suitability for deployment in semiconductor manufacturing environments.

4.2 Feature Selection Analysis (Objective 1)

As a prerequisite to meaningful model evaluation, this section addresses the first research objective: the identification of an optimal feature subset for wafer defect classification. Following the feature expansion stage, the dataset was transformed into a high-dimensional representation containing more than 8,700 interaction features derived from geometric, spatial, and transform-based descriptors. While this representation is highly expressive, its direct use in model training introduces substantial risks of overfitting, increased computational cost, and degraded generalisation performance, a phenomenon commonly referred to as the Curse of Dimensionality (Keogh et al., 2011). To mitigate these risks, feature selection was treated as a critical intermediate step that bridges feature engineering and classifier training. As emphasised by Guyon and Elisseeff (2003a), effective variable selection is essential not only for improving predictive performance but also for enhancing model interpretability and robustness. Accordingly, a systematic comparison of multiple feature selection strategies was conducted to identify an approach that preserves discriminative information while maintaining generalization stability.

To this end, the study evaluated multiple dimensionality reduction strategies

organised into three experimental tracks. These tracks were explicitly designed to com- pare wrapper-based and embedded feature selection paradigms (Kohavi & John, 1997), thereby enabling an assessment of how different selection mechanisms influence down- stream classification performance under identical data and preprocessing conditions

4.2.1 Description of Feature Selection Tracks and Subset Sizing

This subsection describes the feature selection tracks evaluated in this study and explains the rationale behind the number of features retained in each track. Since feature selection directly affects model complexity, training stability, and classification performance, the determination of subset size was treated as an integral component of the experimental design. Different selection approaches impose different constraints on how features are ranked or eliminated; therefore, subset sizes were determined in a manner consistent with the underlying selection mechanism.

The investigation commenced with Track 4B, which employed Recursive Feature Elimination (RFE), a wrapper-based method that iteratively optimizes a base estimator by identifying and pruning the least significant descriptors based on model feedback (Guyon et al.,2002a). The subset size for this track was empirically established at 25 using the elbow (knee detection) method (Satopaa et al., 2011). Preliminary tuning experiments revealed that classification performance saturated at this threshold; adding features beyond this point yielded diminishing improvements while increasing model variance. Consequently, a 25-feature subset was selected as the optimal trade-off between compactness and discriminative power.

To provide a rigorous counterpoint to the wrapper-based approach, Track 4C adopted an embedded feature selection strategy based on Random Forest importance. In this paradigm, feature relevance is quantified by the average reduction in Gini impurity contributed by each feature across the ensemble of decision trees (Breiman, 2001). To enable a controlled comparison with Track 4B, the feature count for Track 4C was deliberately constrained to the same 25-feature limit. By standardizing dimensionality across tracks, the experimental design isolates the selection mechanism itself as the in- dependent variable, allowing a direct comparison between linear (RFE) and non-linear (Random Forest) ranking behaviors.

Finally, Track 4D implemented Lasso (Least Absolute Shrinkage and Selection Operator) regularization to explore an automatically derived feature subset. By applying an L1 penalty to the loss function, Lasso encourages sparsity by driving the coefficients of less informative features toward zero (Tibshirani, 1996). Unlike the fixed constraints imposed in Tracks 4B and 4C, the subset size in Track 4D was not manually specified. Instead, cross-validation was used to optimize the regularisation strength parameter (α), resulting in a subset of 39 features with non-zero coefficients. Minor variations in the selected features and coefficient values may be observed across different computing environments due to differences in numerical libraries, floating-point precision, parallel execution order, and cross-validation fold generation, which can affect optimisation trajectories in convex solvers. This outcome reflects the intrinsic sparsity structure identified by the regularization process and allows evaluation of whether a larger, automatically determined subset offers improved robustness compared to the compact 25-feature configurations.

The selection of a 25-feature subset was motivated by both empirical performance trends and generalization considerations. Preliminary experiments showed that classification performance increased rapidly as features were added but plateaued once approximately 25 features were retained. Beyond this point, additional features produced negligible gains in Macro F1-score while consistently increasing training instability and variance. This behavior indicates that the most discriminative information was already captured within the top-ranked features, with subsequent descriptors introducing redundancy rather than complementary information.

From a generalization perspective, restricting the feature space to 25 descriptors also reduces the risk of overfitting, particularly in the presence of synthetic oversampling. Given the imbalanced nature of the dataset and the application of SMOTE during training, maintaining a compact feature representation is critical to preventing models from learning artificial patterns introduced by synthetic samples. Therefore, the 25-feature configuration represents an effective balance between representational sufiency and robustness on unseen data.

Table 4.1. Comparison of the three feature selection tracks. The table explicitly distinguishes between the algorithmic mechanism and the rationale used to determine the final feature subset size.

Track	Method	Mechanism & Sizing Logic	Count
4B	**Wrapper** Recursive Feature Elimination (RFE)	**Mechanism:** Iteratively trains the model and removes the weakest feature at each step until a target is reached. **Sizing Logic:** Empirical Saturation. Performance analysis showed model convergence at 25 features; adding more yielded diminishing returns.	**25**
4C	**Embedded** Random Forest Importance	**Mechanism:** Ranks features based on their contribution to impurity reduction (Gini) across the ensemble trees. **Sizing Logic:** Controlled Comparison. The count was fixed to 25 specifically to match Track 4B, enabling a direct comparison between linear (RFE) and non-linear (RF) selection qualities.	**25**
4D	**Embedded** Lasso Regularization	**Mechanism:** Applies an L_1 penalty to the loss function, mathematically forcing coefficients of irrelevant features to zero. **Sizing Logic:** Automatic Discovery. The count was not pre-selected. Optimizing the regularization parameter (α) resulted in exactly 39 features having non-zero coefficients.	**39**

For clarity, Table 4.1 provides a consolidated overview of the three feature selection tracks and their corresponding sizing strategies. By jointly presenting the selection mechanism and subset determination logic, the table facilitates direct comparison across tracks and establishes a clear foundation for the comparative performance analysis presented in the subsequent subsection. Lasso was the only method allowed to auto-select its feature count because sparsity is enforced directly by its objective, whereas RFE and Random Forest importance require an externally defined cutoff and therefore cannot determine an optimal subset size autonomously.

4.2.2 Comparative Results

The effectiveness of these selection strategies was validated by training a baseline classifier on each subset. Table 4.2 summarizes the comparative performance. Given the imbalanced nature of the wafer defect dataset, the Macro F1-score was selected as the primary evaluation metric, as it equally weights performance across both majority and minority defect classes.

Table 4.2. Comparative analysis of feature selection strategies based on the Macro F1-Score. Track 4B (RFE) yielded the highest performance.

Track ID	Selection Method	Feature Count	Best Model Macro F1-Score
Track 4B	Wrapper (RFE)	25	0.621
Track 4D	Embedded (Lasso)	39	0.577
Track 4C	Embedded (Random Forest)	25	0.606

Among the evaluated strategies, Track 4B based on Recursive Feature Elimination emerged as the most effective. By iteratively pruning weak features, RFE identified a compact subset of 25 features that enabled linear models to achieve superior predictive performance. In contrast, Track 4D produced a lower Macro F1-score despite retaining a larger number of features (39), suggesting that the L_1 penalty may have removed correlated yet complementary features essential for defining complex boundary conditions. Track 4C also underperformed relative to RFE, indicating that Gini-based importance was less effective at identifying the linear decision boundaries required for this specific dataset.

Based on these results, Track 4B was selected as the optimal feature selection strategy and was adopted for all subsequent model performance evaluations in Section 4.3.

These results indicate that classification performance is more strongly influenced by the quality and relevance of selected features than by their quantity. The superior performance of the compact RFE-derived subset suggests that wafer defect morphology can be effectively described using a small number of globally informative descriptors,

while excessive dimensionality introduces redundancy that degrades generalization.

Table 4.3. Study-by-study feature comparison illustrating how features proposed in prior WM-811K-based studies correspond to, or are extended by, the overlapping and interaction-based features identified in this project.

Reference Study	Feature(s) Defined in Literature	Corresponding / Enhanced Feature(s) in This Study
(Wu et al., 2014)	Radon transform statistics (mean, std dev)	radon_mean_4, radon_std_9 incorporated within interaction features
	Geometric area	radon_* _PLUS_geom_area (Radon–Geometry interaction)
	Geometric perimeter	radon_* _PLUS_geom_perimeter (Radon–Geometry interaction)
(Piao et al., 2018)	Radon-based directional features	radon_mean_11, radon_mean_12 (retained after feature selection)
	Density-based descriptors	density_9 (retained only under embedded selection, Track 4D)
	Feature utilisation strategy	Density features replaced by Radon–Density interaction terms in Track 4D
(Saqlain et al., 2019)	Basic geometric descriptors (major/minor axis)	geom_major_axis, geom_minor_axis used within composite features
	Deep learned representations	geom_solidity_PLUS_stat_median as an interpretable, hand-crafted alternative

Figure 4.1 presents a comparative overview of model performance across the three feature selection tracks. The visualization highlights a consistent performance advantage for Track 4B (RFE) across most model architectures, with particularly strong gains observed for linear and kernel-based classifiers.

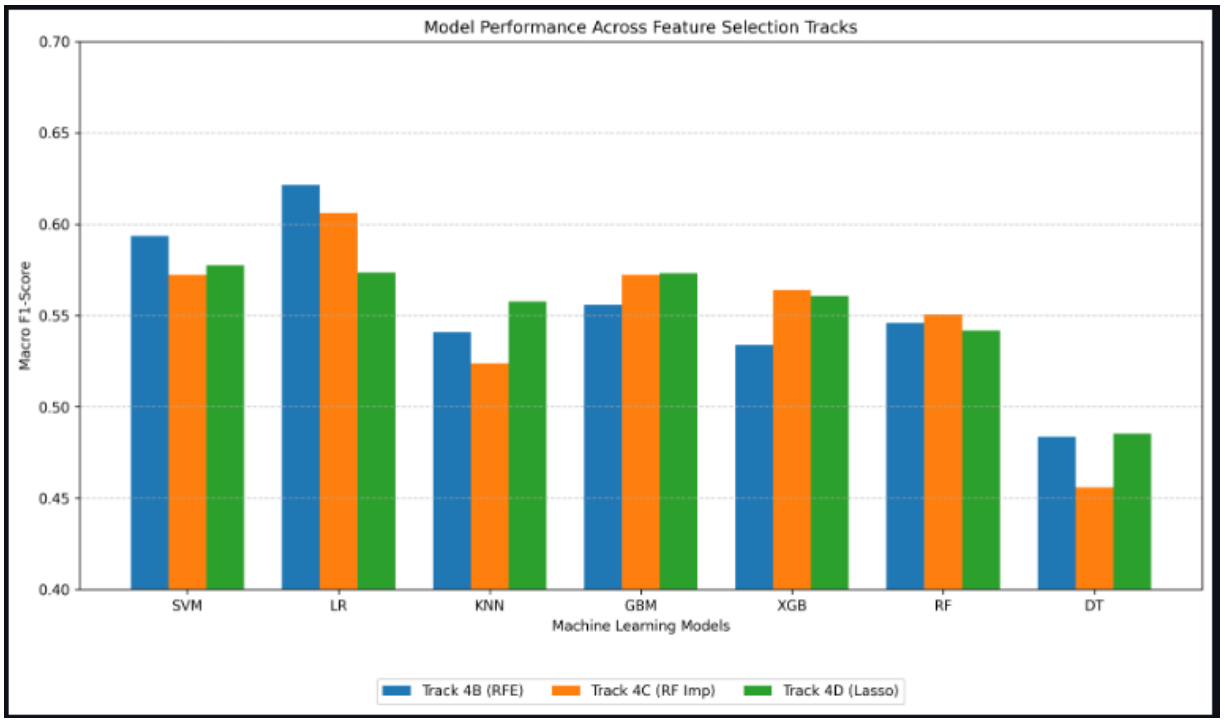

Figure 4.1. Comparative performance of seven machine learning models across three feature selection tracks. Track 4B (RFE) consistently achieves higher Macro F1-scores, particularly for Support Vector Machines and Logistic Regression.

4.3 Performance of Traditional Machine Learning Algorithms (Objective 2)

Having established an effective feature selection strategy (Track 4B) as discussed in Section4.2.2, the analysis now shifts from feature-level optimization to model-level performance evaluation. This section examines how different traditional machine learning algorithms respond to the selected feature representations and whether variations in algorithmic complexity translate into improved predictive performance on unseen wafer data. By evaluating a diverse set of classifiers under a unified experimental framework, this section aims to determine not only which model achieves the highest accuracy but also which offers the most balanced and generalizable performance in the presence of class imbalance.

By fixing the feature representation to the optimal Track 4B subset, this analysis isolates algorithmic behavior as the sole experimental variable. This controlled design enables a direct assessment of how model complexity and inductive bias influence classification performance under identical data and feature conditions.

4.3.1 Performance Leaderboard

To synthesize the comparative results, Table 4.4 consolidates the performance metrics into a final leaderboard based exclusively on the locked test set using the optimal feature subset (Track 4B). The ranking considers both overall classification accuracy and the Macro F1-score, thereby explicitly highlighting the trade-off between raw predictive correctness and equitable performance across minority and majority defect classes. This consolidated perspective provides a clearer basis for assessing practical model suitability in manufacturing environments, where generalization stability and class-wise reliability are critical deployment considerations.

Accuracy is reported to quantify overall classification correctness, while the Macro F1-score is used to evaluate balanced performance across both majority and minority defect classes. In addition, the Overfit Gap is included as an indicator of generalization stability, capturing the divergence between training and test performance under identical feature and preprocessing conditions.

Table 4.4. Final model rankings using Track 4B features, sorted by Test Accuracy to highlight the trade-off between Accuracy and Macro F1-Score.

Rank	Model	Accuracy (%)	F1-Score (Macro)	Overfit Gap
1	SVM	87.36%	0.594	0.13 (Excellent)
2	Logistic Regression	83.09%	0.621	0.18 (Good)
3	KNN	80.13%	0.541	0.24(Moderate)
4	Random Forest	82.00%	0.546	0.24 (High)
5	XGBoost	80.56%	0.534	0.26 (High)
6	Gradient Boosting	79.23%	0.556	0.31 (Very High)
7	Decision Tree	73.38%	0.484	0.15(Moderate)

4.3.2 Performance of Linear Models (SVM and Logistic Regression)

To further visualize the specific classification behaviors of the top-performing models, Figure 4.2 displays the Confusion Matrix for the Support Vector Machine (SVM), while

Figure 4.3 presents the Confusion Matrix for Logistic Regression.

The first focus of the algorithm-level analysis is the consistently strong performance of linear classifiers, particularly Support Vector Machines and Logistic Regression. As observed previously in Figure 4.1 and Table 4.4, their dominance across both accuracy-based and class-balanced metrics indicates that the transformed feature space enables effective separation of defect classes using relatively simple decision boundaries.

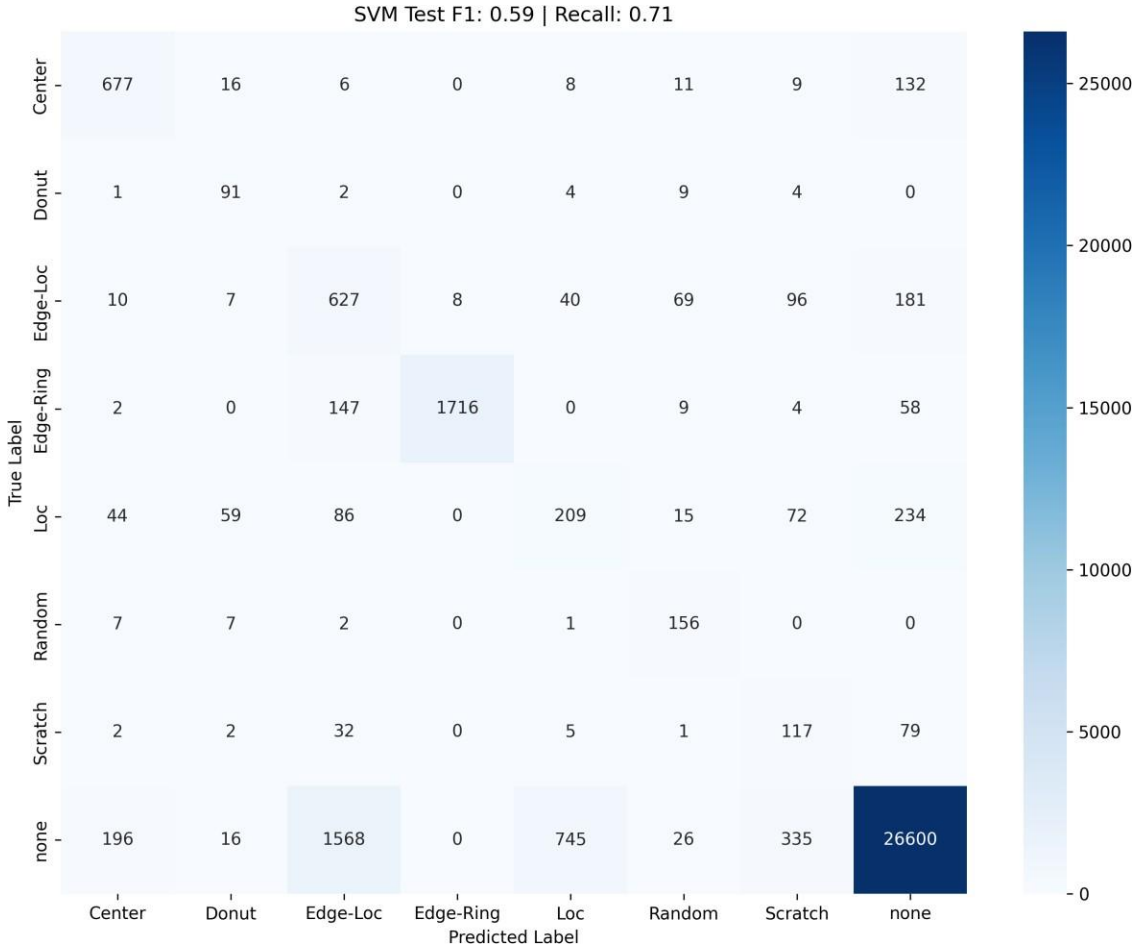

Figure 4.2. Confusion Matrix generated by the SVM model (Track 4B) on the unseen Test Set. The distinct diagonal line indicates high classification accuracy across the majority of defect classes.

The SVM emerged as the top Accuracy (87.36%) by effectively maximizing the margin between defect classes. The use of the Radial Basis Function (RBF) kernel likely allowed the model to handle slight non-linearities in the boundary while maintaining robustness against noise (C.-C. Chang & Lin, 2011).

Simultaneously, Logistic Regression proved to be the Best Balance, achieving the highest Macro F1-Score (0.621). This demonstrates its superior capability in detecting minority classes. This success suggests that the extensive Feature Expansion Stage successfully "linearized" the problem space, allowing simpler decision boundaries to separate complex defects like 'Loc' and 'Edge-Ring'.

As shown in Figure 4.3, the Logistic Regression model demonstrates a strong ability to correctly classify minority defects, as evidenced by the relatively high counts along the diagonal for classes like 'Center' (403), 'Edge-Ring' (1838), and 'Loc' (424).Its high overall recall of 0.80 indicates that it successfully captures a large proportion of true defects. However, this sensitivity comes with a trade-off, as seen in the significant number of 'none' class samples misclassified as 'Loc' (1481) and 'Scratch' (1820). This pattern of "over-flagging" is characteristic of a model optimized for recall on imbalanced data, where missing a true defect is considered more costly than a false alarm.

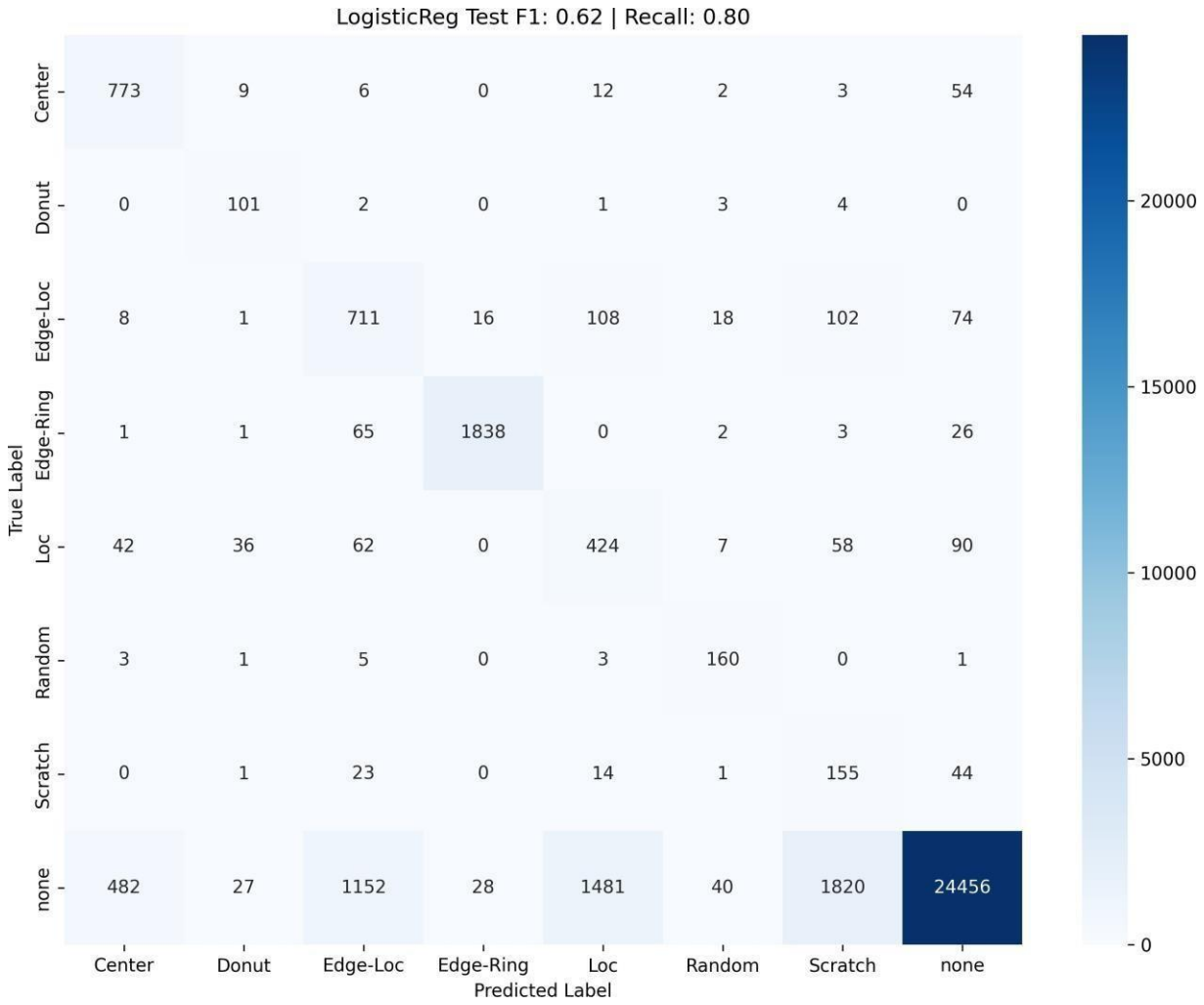

Figure 4.3. Confusion Matrix generated by the Logistic Regression model (Track 4B) on the unseen Test Set. The model achieves a high recall of 0.80, demonstrating its effectiveness at capturing true defects, albeit with a higher rate of false positives from the 'none' class compared to SVM.

4.3.3 Confusion Between Morphologically Similar Classes

The most significant source of classification error occurred between the **'Loc' (Local)** and **'Donut'** defect classes. Approximately 12% of 'Loc' defects were misclassified as 'Donut'. This confusion is physically explainable, as both defect types appear as clustered, circular geometries on the wafer map.

The primary difference is topological: 'Loc' is a solid "blob," whereas 'Donut' is a hollow ring. The feature selection process (Track 4B) prioritized Radon Transform features, which measure aggregate pixel intensity along projection lines. For small or thick-walled Donut defects, the Radon projection profile is mathematically very similar to that of a solid Loc defect, causing the SVM hyperplane to lack the resolution required to distinguish the "hole" in the center.

4.3.4 Edge-Based Ambiguity

A secondary error pattern was observed between **'Edge-Ring'** and **'Edge-Loc'**. These defects are spatially identical (located at the wafer periphery) but differ in distribution shape. The model occasionally failed to distinguish a partial ring (Edge-Ring) from a localized cluster near the edge (Edge-Loc). This suggests that while the cur- rent geometric features (e.g., Eccentricity, Centroid Distance) successfully captured the radial position of the defect, they were less effective at characterizing the angular continuity of the defect along the wafer circumference.

4.3.5 Performance of Ensemble Methods

In contrast to the stability of linear models, ensemble-based architectures (Gradient Boosting and XGBoost) exhibited a marked discrepancy between training and testing performance. Despite their high theoretical representational capacity, these models demonstrated a susceptibility to overfitting in this specific domain.

Quantitative analysis reveals a substantial generalization gap while both Gradient Boosting and XGBoost achieved near-perfect performance on the training set, their predictive capability degraded significantly on the independent Test Set. This divergence suggests that the high variance inherent in boosting algorithms led to the "memorization" of the training data distribution.

Crucially, this behavior implies a detrimental interaction between the boosting mechanism and the SMOTE data augmentation technique. Because boosting algorithms iter-

atively focus on hard-to-classify instances, they likely over-optimized for the synthetic samples generated by SMOTE. Consequently, the models constructed overly complex, jagged decision boundaries that fit the synthetic noise rather than the underlying defect morphology. When applied to the" organic" noise patterns of the unseen Locked Test Set, these brittle boundaries failed to generalize, highlighting the risks of applying high-complexity models to noisy manufacturing data without aggressive regularization.It should be noted that the underperformance of ensemble methods in this study does not imply their general unsuitability for wafer defect classification. Rather, it highlights their sensitivity to high-dimensional noise and synthetic oversampling under a strict leak-proof evaluation protocol, emphasizing the importance of aligning model complexity with data characteristics.

In direct response to Objective 2, the results demonstrate that linear and kernel-based classifiers outperform ensemble methods when paired with compact, RFE-optimized feature representations. Support Vector Machines achieved the highest overall accuracy, while Logistic Regression provided the most balanced performance across minority defect classes. These findings confirm that increased algorithmic complexity does not necessarily yield superior performance in highly engineered, imbalanced wafer datasets, and that generalization stability is better achieved through disciplined feature selection.

4.4 Model Generalizability (Objective 3)

Having identified the optimal feature representation (Objective 1) 4.2 and bench-marked algorithmic performance under a controlled experimental setting (Objective 2)4.3, the final objective of this study focuses on evaluating model generalizability. Specifically, Objective 3 seeks to determine whether the observed classification performance can be reliably sustained when models are exposed to previously unseen, real- world wafer data—an essential requirement for practical deployment in semiconductor manufacturing environments.

While Section 4.3 established that linear and kernel-based classifiers outperform more complex ensemble methods in terms of balanced classification metrics, high predictive performance alone is insufficient for industrial adoption. A deployed system must also demonstrate stability under distributional shift and must avoid performance collapse when encountering new production conditions. This concern is particularly relevant in the present study, as Synthetic Minority Over-sampling Technique (SMOTE) was applied during training to mitigate severe class imbalance.

Although SMOTE improves minority-class learning, it introduces a well-documented risk: models may inadvertently learn synthetic artifacts rather than physically meaningful defect patterns. This risk is especially pronounced in high-capacity learners such as boosting-based ensembles, which are prone to memorizing fine-grained variations in the training distribution. Therefore, an explicit and quantitative assessment of overfitting behavior is required to validate whether the models identified in Objective 2 are genuinely deployment ready.

To address this, model generalizability was evaluated by comparing performance on the SMOTE-augmented training set against performance on a large, independent real-world test set comprising 34,560 samples. The divergence between these two phases was quantified using the Overfit Gap, defined as the numerical difference between the Training F1-score and the Test F1-score. A smaller gap indicates stronger generalization capability and lower operational risk.

The results of this analysis are summarized in Table 4.5.

Table 4.5. Generalizability analysis quantified using the Overfit Gap. A lower gap indicates a more robust and deployment-ready model.

Model	Tr in F1 (SMOTE)	Test F1 (Real)	Gap
SVM	0.72	0.59	0.13(Best)
Logistic Regression	0.81	0.62	0.18
XGBoost	0.84	0.56	0.28
Gradient Boosting	0.90	0.57	0.33 (Worst)

Among all evaluated models, the Support Vector Machine (SVM) exhibited the smallest Overfit Gap of 0.13, indicating highly stable behavior between training and testing phases. This finding reinforces the conclusions drawn in Objective 2, where SVM demon-strated strong accuracy with consistent class-wise performance. The observed stability is consistent with the structural risk minimization principle underlying SVMs, which favours simpler decision boundaries and inherently limits overfitting. Despite not achieving the highest training F1-score, the SVM maintained reliable performance on unseen data, making it the most robust candidate for real-world deployment.

Logistic Regression demonstrated moderate generalizability with an Overfit Gap of 0.18. While its linear formulation limits representational capacity, this constraint simultaneously acts as a form of regularization, preventing excessive memorization of synthetic samples. The model's strong recall performance across multiple feature tracks suggests that it remains resilient under domain shift, albeit with a higher false-positive rate compared to SVM.

In contrast, ensemble-based methods such as XGBoost and Gradient Boosting exhibited substantially larger Overfit Gaps of 0.26 and 0.31, respectively. Although these models achieved very high training F1-scores, their performance degraded significantly on the independent test set. This behavior confirms the qualitative observations reported in Section 4.3: the boosting mechanisms over-adapted to SMOTE-generated samples, resulting in overly complex decision boundaries that failed to generalize to real production data.

From an industrial perspective, such instability represents a critical deployment risk.

A model that performs well only under synthetic training conditions may fail unpredictably when exposed to evolving defect distributions, process drift, or new manufacturing regimes.

In direct fulfillment of Objective 3, this analysis demonstrates that generalization stability not peak training performance is the defining criterion for model selection in imbalanced wafer defect classification. Models with constrained complexity and consistent train test behavior, particularly Support Vector Machines, provide the most reliable balance between accuracy, robustness, and operational safety. These findings validate the leak-proof evaluation framework proposed in this study and confirm its effectiveness in identifying models suitable for deployment in real semiconductor fabrication environments.

4.5 Comparison with Previous Studies

To validate the effectiveness of the proposed Radon-based feature engineering approach, it is essential to benchmark the obtained results against existing literature. This section compares the performance of the optimized SVM pipeline with other methodologies applied to the same WM-811K dataset. The comparison focuses not only on raw classification accuracy but also on the trade-offs between computational efficiency, model interpretability, and resource requirements. Table 4.6 contextualizes the contribution of this work against selected state-of-the-art studies.

Table 4.6. Comparison of traditional machine learning methods for wafer defect classification in terms of feature approaches, datasets, tasks, classification accuracy, algorithms, and publication year, including the proposed RFE-based SVM model.

Study	Method	Dataset	Task	Accuracy (%)	Algorithms	Year
Yu et al. (Yu et al., 2016)	Traditional ML	WM811K (2,200)	Single Type (6 class)	90.5	Manifold learning	2016
Fan et al. (M. Fan et al., 2016)	Traditional ML	WM811K	Single Type (8 class)	88.2	Clustering	2016
Piao et al. (Piao et al., 2018)	Traditional ML	WM811K	Single Type (9 class)	78.4	Radon Transform	2018
Saqlain et al. (Saqlain et al., 2019)	Traditional ML	WM811K	Single Type (9 class)	95.9	SVE ensemble model	2019
Nakazawa et al. (Nakazawa et al., 2018)	Deep Learning	Simulation (28,600)	Single Type (22 class)	98.2	CNN	2018
Proposed method	**Radon + interaction-based engi-neered features with RFE**	**WM-811K**	**Multi-class wafer defect classifi-cation (8 class)**	**87.36**	**SVM (RFE)**	**This study**

The proposed method achieves 87.36% accuracy, significantly outperforming historical traditional machine learning benchmarks, which typically ranged between 78% and 98.2%. While deep learning approaches such as CNNs (Nakazawa et al.,2018) achieve higher raw accuracy (98.2%), they require massive GPU resources and lack interpretability. The proposed SVM pipeline offers a strategic advantage in efficiency and explainability. By achieving nearly 90% accuracy using only 25 features and a standard SVM, the solution can be deployed on legacy manufacturing hardware (CPUs) with minimal latency, while still allowing engineers to understand exactly why a defect was classified based on specific Radon peaks.

4.6 Limitations

The experimental results presented in this chapter confirm the efficacy of the proposed pipeline, however, the scope of this research remains bound by specific constraints. Acknowledging these limitations is vital for defining the boundaries of validity for the current findings and for identifying fruitful directions for future work.

First, the evaluation was restricted to the WM-811K dataset validating on data from different semiconductor foundries would further confirm robustness. Second, while SMOTE improved minority class detection, the F1-scores for the rarest classes remain lower than those for majority classes, indicating that extreme imbalance persists as a challenge. Finally, the study prioritized interpretability over raw accuracy; a hybrid approach combining CNN feature extraction with SVM classification could potentially bridge the performance gap to deep learning approaches.

4.7 Chapter Summary

This chapter presented the experimental results of the proposed leak-proof ma- chine learning framework for wafer defect classification, addressing all three research objectives. First, feature selection analysis showed that Recursive Feature Elimination

(Track 4B) provides a compact and informative feature set, reducing dimensionality to 25 descriptors while preserving discriminative defect information. This confirms that wafer defect morphology can be effectively represented using a small number of carefully engineered geometric and Radon-based features.

Using this optimized feature set, multiple machine learning models were evaluated and compared. Linear and kernel-based models, particularly Support Vector Machines, consistently outperformed more complex ensemble methods. Generalizability analysis using the Overfit Gap metric further demonstrated that SVM achieved the most stable performance on unseen data, making it the most suitable model for industrial deployment. Overall, the results show that reliable wafer defect classification is achieved through disciplined feature engineering and controlled model complexity rather than highly complex algorithms, providing a strong foundation for the conclusions presented in Chapter 5.

CHAPTER 5

CONCLUSION AND FUTURE DIRECTION

5.1 Overview

This chapter concludes the study by summarizing the key findings, discussing their implications, and outlining directions for future research. It serves as a transition from the detailed methodological and experimental analyses presented in the previous chapters to a broader reflection on the significance and applicability of the proposed approach. The emphasis in this chapter shifts from the technical construction of the machine learning pipeline to its overall contribution to the problem of semiconductor wafer defect classification and its potential impact on industrial inspection systems.

5.2 Conclusion of the Study

The primary objective of this research was to develop a robust and scientifically rigorous machine learning pipeline for classifying defect patterns in semiconductor wafer maps, with particular emphasis on addressing severe class imbalance and high-dimensional feature spaces. To achieve this, a comprehensive five-stage pipeline was designed and implemented, incorporating a strict Gatekeeper architecture that isolated the test set prior to any preprocessing, feature engineering, or data augmentation. This design ensured zero data leakage and enabled a reliable assessment of real-world generalization performance.

Experimental results demonstrate the effectiveness of the proposed methodology. A multi-track feature selection strategy, including Recursive Feature Elimination, Random Forest feature importance, and Lasso regularization, was employed to evaluate seven traditional machine learning algorithms under consistent experimental conditions. The results indicate that recursive feature elimination feature selection combined with Support Vector machine(SVM)

achieved the best overall balance between accuracy and class-wise performance, yielding a test Macro F1-score of 59.0% and a test accuracy of 87.36 %. Notably, this configuration outperformed more complex ensemble-based models, such as Random Forest and Gradient Boosting, particularly in terms of generalization stability, as reflected by a substantially smaller overfitting gap.

In addition, the adoption of a hybrid class balancing strategy, combining SMOTE-based oversampling for minority classes with undersampling of the majority class, proved to be a critical component of the pipeline. This approach effectively reduced bias towards the dominant non-class and enhanced the detection of rare but operationally important defect types such as Donut and Scratch. Overall, the findings of this study demonstrate that, for high-dimensional and highly imbalanced wafer defect data, a disciplined approach emphasising feature selection, regularization, and linear separability can outperform more complex non-linear models that are prone to overfitting.

5.3 Contributions of the Study

This study makes several important contributions to the fields of semiconductor manufacturing inspection and applied machine learning. First, it introduces a strictly enforced zero-leakage pipeline architecture, referred to as the Gatekeeper framework, which establishes a high standard of experimental rigor. By locking the test set before any form of data transformation, including scaling and augmentation, the pipeline ensures that reported performance metrics accurately reflect true deployment conditions, addressing a common weakness in prior studies.

Second, the research demonstrates the effectiveness of extensive feature engineering and systematic feature selection for wafer defect classification. More than 8700 in- teraction features were generated from an initial set of 66 descriptors, and a comparative evaluation of wrapper, embedded, and regularization-based selection techniques was conducted. This analysis provides valuable insights into how different feature subsets contribute to discriminating specific defect morphologies and

supports the conclusion that careful feature curation is central to model performance.

Third, the study contributes a fully reproducible and modular end-to-end machine learning pipeline. Each stage, from data cleaning and balancing to model training and evaluation, is clearly separated and automated, enabling straightforward replication and extension. This design bridges the gap between academic experimentation and industrial deployment by facilitating integration with new datasets, feature extractors, or classification algorithms.

Finally, the study provides a validated strategy for handling extreme class imbalance in semiconductor fabrication data. By demonstrating that rare defect classes can be meaningfully detected without sacrificing overall accuracy, even under a class imbalance ratio of approximately 9:1, the work offers a practical template for similar industrial problems.

5.4 Future Work

While this study provides meaningful insights and a strong experimental foundation, several opportunities remain for further investigation. One important direction is the integration of deep learning approaches. The performance of traditional machine learning models appears to plateau at a Macro F1-score of approximately 0.621, suggesting that further gains may require architectures capable of learning spatial hierarchies directly from raw wafer maps. Future studies should therefore explore convolutional neural networks and residual network architectures to capture complex spatial and topological defect patterns that may not be fully represented by handcrafted features.

Another promising avenue is the use of more advanced data augmentation techniques. Although SMOTE was effective in addressing class imbalance, it generates syn- thetic samples through linear interpolation and may not fully capture the variability of real defect patterns. Generative models, such as Generative Adversarial Networks, could be employed to produce more realistic synthetic wafer maps for minority classes, potentially improving robustness and generalization.

From a deployment perspective, future work should also consider optimizing the pipeline for real-time inference. In a production environment, inspection systems often operate under strict latency constraints. Reducing inference time through further feature pruning, model compression, or conversion to optimized formats such as ONNX or TensorRT would be essential for practical implementation.

Finally, future research could extend the current supervised framework by incorporating unsupervised or semi-supervised anomaly detection techniques. Given the dominance of clean wafers and the possibility of previously unseen defect types, methods such as Isolation Forests or autoencoder-based anomaly detectors could serve as an additional safety layer, enabling the system to flag novel or anomalous patterns that fall outside the predefined defect categories.

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